

Principles of Quantum Mechanics – R. Shankar

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1 Mathematical Introduction

1.1 The Delta function

In discrete space with basis $|x_i\rangle$,

- Orthogonality: $\langle x_i | x_j \rangle = \delta_{ij}$
- Completeness: $\sum_{i=1}^n |x_i\rangle \langle x_i| = I$

Natural generalization to continuous functions.

- Orthogonality: $\langle x | x' \rangle = \delta(x - x')$
- Completeness: $\int_a^b |x\rangle \langle x| dx = I$

δ is the Dirac delta function, defined as

$$\begin{aligned} \delta(x - x') &= 0, \quad x \neq x' \\ \int_a^b \delta(x - x') dx' &= 1, \quad a < x < b \end{aligned}$$

Sometimes we write $\delta(0) = \infty$, but this isn't well defined except in the context of an integral. The δ also works like a "sampling function":

$$\int \delta(x - x') f(x') dx' = f(x)$$

Derivative: The derivative of the delta function acts like an operator

$$\delta'(x - x') = \delta(x - x') \frac{d}{dx}$$

To see this, note that

$$\int \delta'(x - x') f(x') dx' = \frac{d}{dx} \int \delta(x - x') f(x') dx' = \frac{df(x)}{dx}$$

Fourier Series: We can also represent the delta function using a fourier series. Recall that

$$\text{Fourier transform: } f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

$$\text{Inverse transform: } f(x') = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx'} f(k) dk$$

$$f(x') = \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk \right) f(x) dx$$

Reasoning from the use of delta as a sampling function,

$$\delta(x' - x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x'-x)} dk$$

1.2 The Derivative Operator

We can have operators in infinite dimensions, e.g.

$$\Omega |f\rangle = |\tilde{f}\rangle$$

The derivative is one such operator:

$$D |f\rangle = \left| \frac{df}{dx} \right\rangle$$

The elements of D in the x basis are

$$D_{xx'} \equiv \langle x | D | x' \rangle = \delta'(x - x') = \delta(x - x') \frac{d}{dx'}$$

To see this, recall the use of δ as a sampling function.

$$\begin{aligned} \langle x | D | f \rangle &= \langle x | \frac{df}{dx} \rangle = \frac{df(x)}{dx} \\ \int \langle x | D | x' \rangle \langle x' | f \rangle dx' &= \frac{df}{dx} \end{aligned}$$

1.2.1 The Hermitian Operator $K = -iD$

However, D is not Hermitian: $D_{xx'} \neq D_{x'x}^*$.

$$\begin{aligned} D_{xx'} &= \delta'(x - x') \\ D_{x'x}^* &= \delta'(x' - x)^* = -\delta'(x - x') \end{aligned}$$

Instead, we can make D Hermitian using a constant complex term: $K = -iD$. Then

$$\begin{aligned} K_{xx'} &= -i\delta'(x - x') \\ K_{x'x}^* &= [-i\delta'(x' - x)]^* = i\delta'(x' - x) = -i\delta(x - x') \end{aligned}$$

Warning: For operators in infinite dimensions, $K_{xx'} = K_{x'x}^*$ doesn't actually guarantee Hermiticity. To see this, note that a Hermitian operator must satisfy (for functions f, g)

$$\begin{aligned}\langle g|K|f\rangle &= \langle f|K|g\rangle^* \\ \int_a^b \int_a^b \langle g|x\rangle \langle x|K|x'\rangle \langle x'|f\rangle dx dx' &\stackrel{?}{=} \left(\int_a^b \int_a^b \langle f|x\rangle \langle x|K|x'\rangle \langle x'|g\rangle dx dx' \right)^* \\ \int_a^b g^*(x) \left[-i \frac{df(x)}{dx} \right] dx &\stackrel{?}{=} \left(\int_a^b f^*(x) \left[-i \frac{dg(x)}{dx} \right] dx \right)^* \\ -ig^*(x)f(x) \Big|_a^b + i \int_a^b \frac{dg^*(x)}{dx} f(x) dx &\neq i \int_a^b \frac{dg^*(x)}{dx} f(x) dx\end{aligned}$$

In order to be Hermitian, the **surface term** of K must be 0. K is Hermitian in the space of functions where this holds. $K_{xx'} = K_{x'x}^*$ is no longer a sufficient condition.

$$-ig^*(x)f(x) \Big|_a^b = 0$$

Examples where

- String clamped at one end, $f(0) = f(L) = 0$. Derivative is $K = -i \frac{d}{dx}$
- Periodic function, $f(\phi) = f(\phi + 2\pi)$. Derivative is $K = -i \frac{d}{d\phi}$

In QM, we study functions $-\infty < x < \infty$. There are two classes of such functions.

1. f vanishes as $x \rightarrow -\infty$ and $x \rightarrow \infty$. Here, $K = -i \frac{d}{dx}$ is Hermitian.
2. f doesn't vanish, like e^{ikx} . If K is Hermitian, it must satisfy

$$e^{ikx} e^{ik'x} \Big|_{-\infty}^{\infty} = 0$$

If $k = k'$, this holds. Otherwise, we define the limit as $|x| \rightarrow \infty$ to be the average over a large interval. Then if $k \neq k'$,

$$\lim_{x \rightarrow \infty} e^{ikx} e^{ik'x} = \lim_{L, \Delta \rightarrow \infty} \frac{1}{\Delta} \int_L^{L+\Delta} e^{i(k-k')x} dx = 0$$

Thus K is also Hermitian for class 2 functions.

1.2.2 Eigenvalues of K

For eigenvalue k ,

$$\begin{aligned}K|k\rangle &= k|k\rangle \\ \langle x|K|k\rangle &= k\langle x|k\rangle \\ \int \langle x|K|x'\rangle \langle x'|k\rangle dx' &= k\langle x|k\rangle\end{aligned}$$

Let $\langle x|k\rangle = \psi_k(x)$. Using the definition $K = -iD$,

$$-i \frac{d}{dx} \psi_k(x) = k \psi_k(x)$$

The solution is $\psi_k(x) = A e^{ikx}$. Thus, any real number k is an eigenvalue of K with eigenvector $A e^{ikx}$. Since A is a free parameter, set it to $1/\sqrt{2\pi}$ so that

$$|k\rangle \leftrightarrow \frac{1}{\sqrt{2\pi}} e^{ikx}$$

$$\langle k|k'\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|k'\rangle dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx = \delta(k' - k)$$

Thus $\langle k|k'\rangle = 0$ if $k \neq k'$, and blows up if $k = k'$.

We require a real eigenvalue k because if k is imaginary, e^{ikx} blows up as $x \rightarrow \infty$. The choice of a real eigenvalue restricts us to the **physical Hilbert space**.

1.2.3 Conjugate actions of X and K

In finite dimensions, the eigenfunctions of a Hermitian operator form a complete basis. Assume this holds for infinite dimensions.

K is Hermitian. Expand in both X and K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$f(k) = \langle k|f\rangle = \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|f\rangle dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} f(x) dx$$

$$f(x) = \langle x|f\rangle = \int_{-\infty}^{\infty} \langle x|k\rangle \langle k|f\rangle dk = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ikx} f(k) dk$$

The Fourier transform pass f from one complete basis to another.

Additionally, note the matrix elements of X and K in the X and K basis, respectively:

$$\langle x|X|x'\rangle = x' \delta(x - x')$$

$$\langle k|K|k'\rangle = k' \delta(k - k')$$

In finite dimensions, we had (in the X basis) $X|x\rangle = x|x\rangle$. To see the action of X in infinite dimensions, let $X|f\rangle = |\tilde{f}\rangle$.

$$\langle x|X|f\rangle = \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|f\rangle dx' = x f(x)$$

$$\langle x|X|f\rangle = \langle x|\tilde{f}\rangle = \tilde{f}(x)$$

Thus $\tilde{f}(x) = x f(x)$, and the effect of X in infinite dimensions is $X|f(x)\rangle = |x f(x)\rangle$. That is, multiplying $f(x)$ by x .

An interesting phenomenon occurs if we calculate the matrix elements of X in the K basis. Recall $|k\rangle = \frac{1}{\sqrt{2\pi}} e^{ikx}$.

$$\begin{aligned}\langle k|X|k'\rangle &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \langle k|x\rangle \langle x|X|x'\rangle \langle x'|k'\rangle dx dx' \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x' \delta(x-x') \langle k|x\rangle \langle x'|k'\rangle dx dx' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} e^{ik'x} dx \\ &= i \frac{d}{dk} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(k'-k)x} dx \right) = i \delta'(k-k')\end{aligned}$$

Thus, if g is a function with image $g(k)$ in the K basis,

$$X |g(k)\rangle = |i \frac{dg(k)}{dk}\rangle$$

To summarize,

$$\text{In } X \text{ basis } \begin{cases} X \equiv x \\ K \equiv -i \frac{d}{dx} \end{cases}$$

$$\text{In } K \text{ basis } \begin{cases} X \equiv i \frac{d}{dx} \\ K \equiv k \end{cases}$$

Operators like these are **conjugate** to each other.

Note that X and K do not commute.

$$\begin{aligned}XK|f\rangle &= -ix \frac{df(x)}{dx} \\ KX|f\rangle &= -i \frac{d}{dx} (xf(x)) = -ix \frac{df(x)}{dx} - if(x)\end{aligned}$$

Thus $[X, K]|f\rangle = if = iI|f\rangle$. Since f was arbitrary, we have $[X, K] = iI$.

In QM, X and K are the main operators we'll be seeing (X gives position, K gives derivative). However, they take different forms: $\hat{X} = X$ and $\hat{P} = \hbar K$.

2 Review of Classical Mechanics

2.1 Lagrangian Mechanics

Newtonian mechanics can find the path of a particle at an infinitesimal time slice:

$$m\ddot{x} = -\frac{dV}{dx}$$

Given $x(t_i)$ and $\dot{x}(t_i)$ at a point t_i , we can obtain the classical path's $x_{\text{cl}}(t_i + \Delta t)$ and $\dot{x}_{\text{cl}}(t_i + \Delta t)$ after some infinitesimal time Δt using the equation above. By continuing this process, we recreate the classical path x_{cl} of the particle.

$$\begin{aligned}x_{\text{cl}}(t_i + \Delta t) &= x(t_i) + \dot{x}(t_i)\Delta t \\ \dot{x}_{\text{cl}}(t_i + \Delta t) &= \dot{x}(t_i) + \ddot{x}(t_i)\Delta t\end{aligned}$$

Lagrangian mechanics seeks to find the classical path immediately. Given two points (x_i, t_i) and (x_f, t_f) , we can find the path taken by the particle between them. But what distinguishes the classical path from all other possible ones?

2.1.1 Procedure

1. Define the Lagrangian $L = T - V$. Assume time-independent potential, thus $L = L(x, \dot{x})$.
2. For each path from (x_i, t_i) to (x_f, t_f) , define the action as a function of the entire path:

$$S[x(t)] = \int_{t_i}^{t_f} L(x, \dot{x}) dt$$

3. **Principle of least action:** the classical path is that on which S is minimized.

NB: assumption of a time-independent potential doesn't always hold. For example, $F_B = qv \times B$. So $L = T - V$ isn't always the correct form.

2.1.2 The Euler-Lagrange Equation

For finite variables, the function was at a extremal if the 1st order change δf vanishes:

$$\begin{aligned}f(\mathbf{x}^0 + \eta) &= f(\mathbf{x}^0) + \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i + O(\eta^2) \\ \delta f &= \sum_{i=1}^n \left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}^0} \eta_i = 0\end{aligned}$$

Assume this holds for functions of infinite variables, like $S[x(t)]$. Let $x_{\text{cl}}(t)$ be the classical path, and $x_{\text{cl}}(t) + \eta(t)$ a nearby path. Endpoints must match, so $\eta(t_i) = \eta(t_f) = 0$. Now find the extremal where 1st order change δS vanishes:

$$\begin{aligned}S[x_{\text{cl}}(t) + \eta(t)] &= \int_{t_i}^{t_f} L(x_{\text{cl}}(t) + \eta(t), \dot{x}_{\text{cl}}(t) + \dot{\eta}(t)) dt \\ &= \int_{t_i}^{t_f} \left[L(x_{\text{cl}}(t), \dot{x}_{\text{cl}}(t)) + \left. \frac{\partial L}{\partial x(t)} \right|_{x_{\text{cl}}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{\text{cl}}} \cdot \dot{\eta}(t) + \dots \right] dt \\ &\approx S[x_{\text{cl}}(t)] + \delta S\end{aligned}$$

Find where $\delta S = 0$ for an arbitrary deviation η :

$$\begin{aligned}
 0 = \delta S &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) + \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \dot{\eta}(t) \right] dt \\
 &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt + \left[\left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right]_{t_i}^{t_f} - \int_{t_i}^{t_f} \left[\frac{d}{dt} \left. \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \cdot \eta(t) \right] dt \\
 &= \int_{t_i}^{t_f} \left[\left. \frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right|_{x_{cl}} \right] \cdot \eta(t) dt
 \end{aligned}$$

Thus we get the Euler-Lagrange equation, which must be satisfied by a classical path:

$$\left[\frac{\partial L}{\partial x(t)} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}(t)} \right]_{x_{cl}} = 0$$

2.1.3 Conservation Laws from the Lagrangian

A benefit of the Lagrangian is that unlike the Newtonian, coordinates don't have to be Cartesian. We can have an arbitrary q_i and $\dot{q}_i$.

Further note:

- Canonical momentum conjugate: $p_i = \partial L / \partial \dot{q}_i$
- Canonical force conjugate: $F_i = \partial L / \partial q_i$

$$F_i = \frac{d}{dt} p_i$$

E.g. if $q_i = \theta$, then p_i is angular momentum and F_i is torque.

Call q_i a **cyclic coordinate** if L depends on $\dot{q}_i$, but not q_i . Then p_i is conserved:

$$\frac{dp_i}{dt} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} = 0$$

Conservation laws fall out naturally out of the Lagrangian, while it may be opaque in the Newtonian. For example, consider a Lagrangian in polar coordinates $L = L(\rho, \phi)$. The metric is $ds^2 = d\rho^2 + \rho d\phi^2$. Thus the infinitesimal velocity is

$$v = \frac{ds}{dt} = \sqrt{(\dot{\rho})^2 + (\rho \dot{\phi})^2}$$

The Lagrangian is

$$L = \frac{1}{2} m [\dot{\rho}^2 + \rho^2 \dot{\phi}^2] - V(\rho, \phi)$$

Applying Euler-Lagrange for ρ and ϕ respectively gives

$$\begin{aligned}
 \frac{d}{dt} (m\dot{\rho}) &= -\frac{\partial V}{\partial \rho} + m\rho\dot{\phi}^2 \quad \longrightarrow \quad m\ddot{\rho} = -\frac{\partial V}{\partial \rho} + m\rho(\dot{\phi})^2 \\
 \frac{d}{dt} (m\rho^2\dot{\phi}) &= -\frac{\partial V}{\partial \phi} \quad \longrightarrow \quad m\ddot{\phi} = -\frac{1}{\rho^2} \frac{\partial V}{\partial \phi} - \frac{1}{\rho} (2m\dot{\rho}\dot{\phi})
 \end{aligned}$$

Note how easily we can derive the centrifugal and coriolis forces. Also note that conservation laws for cyclic coordinates $V(\rho, \phi) = V(\rho)$ and $V(\rho, \phi) = V(\phi)$ fall out naturally, and the corresponding equations of motion are that of $\partial V/\partial \phi = 0$ and $\partial V/\partial \rho = 0$.

In sum:

- Equations of motion are simpler to find with Lagrangian
- Conservation laws are easy to see from Lagrangian
- Path integral formalism uses Lagrangian (later)

2.1.4 The Electromagnetic Lagrangian

Electromagnetic force:

$$\vec{F} = q \left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B} \right)$$

Note: this is Gaussian units, hence the $1/c$ factor for magnetic force.

The electromagnetic force is velocity-dependent, so $L = T - V$ doesn't work. Instead, we construct L to reduce to the expected equations. The EM lagrangian is

$$L_{\text{EM}} = \frac{1}{2} m \vec{v} \cdot \vec{v} - q\phi + \frac{q}{c} \vec{v} \cdot \vec{A}$$

ϕ and $\vec{A}$ are the scalar and vector potentials, related to E and B by

$$\begin{aligned} \vec{E} &= -\nabla\phi - \frac{1}{c} \frac{\partial \vec{A}}{\partial t} \\ \vec{B} &= \nabla \times \vec{A} \end{aligned}$$

The Euler-Lagrange equation is

$$\begin{aligned} \frac{d}{dt} \left(m\dot{x}_i + \frac{q}{c} A_i \right) &= -q \frac{\partial \phi}{\partial x_i} + \frac{q}{c} \frac{\partial (\vec{v} \cdot \vec{A})}{\partial x_i}, \quad i = 1, 2, 3 \\ \frac{d}{dt} \left(m\vec{v} + \frac{q}{c} \vec{A} \right) &= -q \nabla \phi + \frac{q}{c} \nabla (\vec{v} \cdot \vec{A}) \\ \frac{d}{dt} (m\vec{v}) &= -q \nabla \phi + \frac{q}{c} \left[-\frac{d\vec{A}}{dt} + \nabla (\vec{v} \cdot \vec{A}) \right] \end{aligned} \tag{1}$$

By time-dependence of the magnetic vector potential $\vec{A}$,

$$\frac{d\vec{A}}{dt} = \frac{\partial \vec{A}}{\partial t} + \sum_i \frac{\partial \vec{A}}{\partial x_i} \frac{dx_i}{dt} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \nabla) \vec{A}$$

Thus equation (1) becomes

$$\frac{d}{dt} (m\vec{v}) = -q \nabla \phi - \frac{q}{c} \frac{\partial \vec{A}}{\partial t} + \frac{q}{c} \underbrace{\left[\nabla (\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \nabla) \vec{A} \right]}_{\vec{v} \times (\nabla \times \vec{A})} \tag{2}$$

This reduces to $\vec{F} = q\vec{E} + \frac{q}{c}\vec{v} \times \vec{B}$.

The time-dependent EM field is usually not conservative, so there's no path-independent work function to use as the potential. Thus L_{EM} is made to fit the situation and reduce as expected. $L = T - V$ only works for conservative forces.

2.1.5 The Two-Body Problem

Assume 2 masses m_1, m_2 that exert equal and opposite forces on each other. We can reduce this to a one-body problem with Lagrangian mechanics.

The potential only depends on relative position, $V(\vec{r}_1, \vec{r}_2) = V(\vec{r}_1 - \vec{r}_2)$. Thus $6 \rightarrow 3$ cyclic coordinates in 3D.

The two-body Lagrangian is

$$L = \frac{1}{2}m_1|\dot{\vec{r}}_1|^2 + \frac{1}{2}m_2|\dot{\vec{r}}_2|^2 - V(r_1 - r_2) \quad (1)$$

Define $\vec{r}_{\text{CM}}$ and $\vec{r}$ such that

$$\begin{aligned} \vec{r}_{\text{CM}} &= \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}, \quad \vec{r} = \vec{r}_1 - \vec{r}_2 \\ \vec{r}_1 &= \vec{r}_{\text{CM}} + \frac{m_2\vec{r}}{m_1 + m_2} \\ \vec{r}_2 &= \vec{r}_{\text{CM}} - \frac{m_1\vec{r}}{m_1 + m_2} \end{aligned}$$

The two-body Lagrangian becomes

$$L = \frac{1}{2}(m_1 + m_2)|\dot{\vec{r}}_{\text{CM}}|^2 + \frac{1}{2}\frac{m_1m_2}{m_1 + m_2}|\dot{\vec{r}}|^2 - V(\vec{r}) \quad (2)$$

This can be thought of as the Lagrangian of two fictitious particles that don't interact: one "CM" particle with mass $M = m_1 + m_2$, and one "reduced mass" particle with mass $\mu = (m_1m_2)/(m_1 + m_2)$. Note that $L(r, \dot{r}, r_{\text{CM}}, \dot{r}_{\text{CM}}) = L(r, \dot{r}) + L(r_{\text{CM}}, \dot{r}_{\text{CM}})$.

By going to the CM frame, we can eliminate $\dot{r}_{\text{CM}}$ and just solve for the motion of μ . We've reduced the two-body problem to a one-body problem.

2.2 Hamiltonian Mechanics

2.2.1 Derivation

The Lagrangian uses $L(q, \dot{q})$. The Hamiltonian reverses the role of p and $\dot{q}$, using $H(q, p)$. This "role reversal" is achieved with a **Legendre transform**. Suppose $u(x) = df/dx$ and $x(u) = dg/du$. The relationship between f and g is a Legendre transform:

$$g(u) = x(u)u - f(x(u))$$

Define the Hamiltonian $H(q, p)$ as a Legendre transform of the Lagrangian $L(q, \dot{q})$:

$$H(q, p) = \sum_{i=1}^n p_i \dot{q}_i - L(q, \dot{q}) \quad , \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Hamilton's **canonical equations** can be found:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad , \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Lagrangian	Hamiltonian
n coordinates q_i and n velocities $\dot{q}_i$	n coordinates q_i and n momenta p_i
State of system is point in n -dimensional configuration space	State of system is point in $2n$ -dimensional configuration space
n coordinates evolve according to n 2nd-order differential equations	$2n$ coordinates and momenta evolve according to $2n$ 1st-order differential equations
For given L , multiple trajectories can pass through a point in configuration space depending on $\dot{q}_i$.	For given H , only one trajectory passes through a point in configuration space.

For conservative forces with $T = \frac{1}{2}m\dot{x}_i^2$, we get

$$p_i = \frac{\partial L}{\partial \dot{x}_i} = m_i \dot{x}_i$$

$$H = \sum_i p_i \dot{x}_i - L(x, \dot{x}) = T + V$$

But this isn't always the right form. It's only for conservative forces.

2.2.2 The Electromagnetic Hamiltonian

If L_{EM} generates the correct equations of motion, then H_{EM} will as well. But it won't take the form $H = T + V$ since now the force isn't conservative.

Start from the electromagnetic Lagrangian:

$$L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} - q\phi + \frac{q}{c}\vec{v} \cdot \vec{A}$$

Now find the canonical momentum $\vec{p}$ and the Hamiltonian H :

$$\vec{p} = \frac{\partial L}{\partial \vec{v}} = m\vec{v} + \frac{q}{c}\vec{A}$$

$$H_{\text{EM}} = \vec{p} \cdot \vec{v} - L_{\text{EM}} = \frac{1}{2}m\vec{v} \cdot \vec{v} + q\phi = \frac{|\vec{p} - q\vec{A}/c|^2}{2m} + q\phi$$

2.2.3 Conservation Laws from the Hamiltonian

If q_i is a cyclic coordinate in the Hamiltonian, then

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} = 0$$

The associated momentum is conserved. But other quantities are too, and this is often easier to see in the Hamiltonian than in the Lagrangian.

Let $\omega(p, q)$ be a state function with **no explicit dependence on t** . Define the **Poisson bracket**

$$\{\omega, \lambda\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial \lambda}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial \lambda}{\partial q_i} \right)$$

The time derivative of ω is:

$$\begin{aligned} \frac{d\omega}{dt} &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial q_i}{\partial t} + \frac{\partial \omega}{\partial p_i} \frac{\partial p_i}{\partial t} \right) \\ &= \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \\ &= \{\omega, H\} \end{aligned}$$

Any state function ω whose Poisson bracket with H vanishes is conserved. H itself is conserved if there's no explicit t dependence.

The Poisson bracket between q and p satisfy

$$\begin{aligned} \{q_i, q_j\} &= \{p_i, p_j\} = 0 \\ \{q_i, p_j\} &= \delta_{ij} \end{aligned}$$

From these, the canonical equations can be derived in terms of Poisson brackets:

$$\begin{aligned} \dot{q}_i &= \{q_i, H\} \\ \dot{p}_i &= \{p_i, H\} \end{aligned}$$

2.2.4 Canonical Transformations

Euler-Lagrange equations are invariant under a coordinate change $(q, \dot{q}) \rightarrow (\bar{q}, \dot{\bar{q}})$, which implies the invariance of Hamilton's equations under $(q, p) \rightarrow (\bar{q}, \bar{p})$.

General transformations of the form $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ are called **canonical transformations** if they obey the canonical equations. $(\bar{q}, \bar{p})$ are called **canonical coordinates**.

Theorem: If $(\bar{q}(q, p), \bar{p}(q, p))$ are canonical coordinates iff.

$$\begin{aligned} \{\bar{q}_j, \bar{q}_k\} &= 0 = \{\bar{p}_j, \bar{p}_k\} \\ \{\bar{q}_j, \bar{p}_k\} &= \delta_{jk} \end{aligned}$$

Proof: For state functions $\omega(q, p)$,

$$\dot{\omega} = \{\omega, H\} = \sum_i \left(\frac{\partial \omega}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \omega}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$$

Thus for $\dot{q}_j$,

$$\dot{q}_j = \sum_i \left(\frac{\partial \bar{q}_j}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \bar{q}_j}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \quad (1)$$

Using the chain rule, we can also get

$$\begin{cases} \frac{\partial H(q, p)}{\partial p_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial p_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial p_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial p_i} \right) \\ \frac{\partial H(q, p)}{\partial q_i} = \frac{\partial H(\bar{q}, \bar{p})}{\partial q_i} = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \frac{\partial \bar{q}_k}{\partial q_i} + \frac{\partial H}{\partial \bar{p}_k} \frac{\partial \bar{p}_k}{\partial q_i} \right) \end{cases} \quad (2)$$

Plugging (2) into (1) and simplifying gives (following a similar procedure for $\dot{p}_j$),

$$\begin{cases} \dot{q}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{\bar{q}_j, \bar{q}_k\} + \frac{\partial H}{\partial \bar{p}_k} \{\bar{q}_j, \bar{p}_k\} \right) \\ \dot{p}_j = \sum_k \left(\frac{\partial H}{\partial \bar{q}_k} \{\bar{p}_j, \bar{q}_k\} + \frac{\partial H}{\partial \bar{p}_k} \{\bar{p}_j, \bar{p}_k\} \right) \end{cases} \quad (3)$$

For (3) to reduce to the canonical equations, we must have $\{\bar{q}_j, \bar{q}_k\} = 0 = \{\bar{p}_j, \bar{p}_k\}$ and $\{\bar{q}_j, \bar{p}_k\} = \delta_{jk}$.

*Note that the Poisson bracket is independent of parametrization. Whether $\omega(q, p)$ or $\omega(\bar{q}, \bar{p})$,

$$\{\omega, \lambda\}_{q, p} = \{\omega, \lambda\}_{\bar{q}, \bar{p}}$$

2.2.5 Active Transformations

The transformation $q \rightarrow \bar{q}(q, p)$ and $p \rightarrow \bar{p}(q, p)$ is passive if (q, p) and $(\bar{q}, \bar{p})$ refer to the same point in phase space.

If the transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ preserves the range of variables, it's called **regular**.

- E.g. transformation from a Cartesian plane to a translated/rotated one is regular.
- But a transformation from a Cartesian plane to spherical coordinates isn't. Some variables are now bounded by 2π instead of ∞ .

A transformation that "shifts" the phase space so that (q, p) and $(\bar{q}, \bar{p})$ no longer refer to the same point is called **active**. Active transformations are a subset of regular ones.

We call ω invariant under the regular transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$ if

$$\omega(q, p) = \omega(\bar{q}, \bar{p})$$

2.2.6 Symmetries

Two results:

1. If H is invariant under the following infinitesimal transformations, then $g(q, p)$ is conserved. We call g a **generator of the transformation**.

$$\begin{aligned} q_i &\rightarrow \bar{q}_i + \varepsilon \frac{\partial g}{\partial p_i} \equiv q_i + \delta q_i \\ p_i &\rightarrow \bar{p}_i - \varepsilon \frac{\partial g}{\partial q_i} \equiv p_i + \delta p_i \end{aligned}$$

2. If H is invariant under the regular, canonical, but not necessarily infinitesimal transformation $(q, p) \rightarrow (\bar{q}, \bar{p})$, and if (q, p) is a solution to the equations of motion, so is $(\bar{q}, \bar{p})$.

Proof of (1): Note that

$$\begin{aligned} \delta H &= \sum_i \frac{\partial H}{\partial q_i} (\delta q_i) + \frac{\partial H}{\partial p_i} (\delta p_i) \\ &= \sum_i \frac{\partial H}{\partial q_i} \left(\varepsilon \frac{\partial g}{\partial p_i} \right) + \frac{\partial H}{\partial p_i} \left(-\varepsilon \frac{\partial g}{\partial q_i} \right) \\ &= \varepsilon \{H, g\} \end{aligned}$$

For H to be invariant (to first order), $\delta H = 0$ and $\{H, g\} = 0$. Thus g is conserved.

In general for state variables ω ,

$$\delta \omega = \varepsilon \{\omega, g\}$$

2.2.7 A Relation Between S and E

Let x_{cl} be a classical path from (x_i, t_i) to (x_f, t_f) , $H(t_f)$ be the Hamiltonian at t_f , and S the classical action. A neat relation is

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -H(t_f)$$

If E is conserved, this becomes

$$\frac{\partial S_{\text{cl}}(x_f, t_f; x_i, t_i)}{\partial t_f} = -E$$

Proof: Hold x_i, x_f fixed and vary with $\eta(t)$. For convenience, $x_i = t_i = 0$ and $\eta(0) = 0$.

$$x(t) = x_{\text{cl}}(t) + \eta(t)$$

The variation in action can be split into (1) the contribution from difference in path, (2) the contribution from the new “end” time $t = t_f + \Delta t$.

$$\begin{aligned}
\delta S_{\text{cl}} &= \int_0^{t_f} \frac{\partial L}{\partial t} dt + L(t_f)\Delta t \\
&= \int_0^{t_f} \left[\frac{\partial L}{\partial x} \eta(t) + \frac{\partial L}{\partial \dot{x}} \dot{\eta}(t) \right] dt + L(t_f)\Delta t \\
&= \int_0^{t_f} \left[-\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} + \frac{\partial L}{\partial x} \right] \eta(t) dt + \int_0^{t_f} \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right] dt + L(t_f)\Delta t \\
&= 0 + \left[\frac{\partial L}{\partial \dot{x}} \eta(t) \right]_0^{t_f} + L(t_f)\Delta t
\end{aligned}$$

Substitute $\eta(t_f) = -\dot{x}(t_f)\Delta t$ to get

$$\delta S = \left[-\frac{\partial L}{\partial \dot{x}} \dot{x} + L \right]_{t_f} \Delta t = -H(t_f)\Delta t$$

2.3 A Problem: Wave-Particle Duality

2.3.1 Plane Waves

An electromagnetic wave ψ that travels at c obey the wave equation

$$\nabla^2 \psi = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

A special case is the plane wave:

$$\psi(x, t) = Ae^{i\phi} = A \exp \left(i \left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t \right) \right)$$

- Wave number $k = 2\pi/\lambda$ (phase change per unit length at fixed t)
- Angular frequency $\omega = 2\pi/T$ (phase change per unit time at fixed x)
- Wave speed $v = \omega/T$

This can be extended to plane waves in 3D, using the **wave vector** $\vec{k}$.

$$\psi(\vec{r}, t) = Ae^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad , \quad \omega = |\vec{k}|v$$

2.3.2 Double-Slit Experiment

The wave prediction is $I_{1+2} \neq I_1 + I_2$, due to interference effects.

The classical particle prediction is $I_{1+2} = I_1 + I_2$.

By opening one slit and lowering the intensity, we find that light energy comes in bundles. Each bundle has energy and momentum that satisfies

$$E = \hbar\omega \quad , \quad p = \hbar k$$

Thus light seems to be made up of particles called photons. But if both slits are opened at I , we get a wave effect: $I_{1+2} \neq I_1 + I_2$. The difference can't be reconciled classically.

Born suggested that each photon is associated with a wave ψ called the **probability amplitude**. $|\psi(x)|^2$ is the probability of finding the particle at x . The wave is described by

$$\omega = \frac{E}{\hbar} \quad , \quad k = \frac{p}{\hbar}$$

The wave interferes with itself (after being split by the double-slit) to form the overall pattern according to $|\psi(x)|^2$.

De Broglie suggested that every particle of momentum p has a wave associated with it, not just massless photons. The wave number is

$$k = \frac{p}{\hbar}$$

3 Postulates of Quantum Mechanics

3.1 The Postulates

Classical:

1. State of particle is described by $x(t)$ and $p(t)$ in 2D phase space.
2. Every dynamical variable $\omega(x, p)$ is a function of x, p .
3. If a particle is in the state $(x; p)$, a perfect measurement of ω yields $\omega(x, p)$ without changing the state.
4. Time evolution described by

$$\dot{x} = \frac{\partial H}{\partial p} \quad , \quad \dot{p} = -\frac{\partial H}{\partial x}$$

Quantum:

1. State of particle is a vector $|\psi(t)\rangle$ in Hilbert space.
2. Independent variables x, p of classical mechanics become Hermitian operators X, P such that

$$\begin{aligned} \langle x|X|x'\rangle &= x\delta(x-x') \\ \langle x|P|x'\rangle &= -\hbar\delta'(x-x') \end{aligned}$$

(Note that this is just X and $P = \hbar K$ from before). Then $\omega(x, p)$ becomes

$$\Omega(X, P) = \omega(x \rightarrow X, p \rightarrow P)$$

3. If a particle is in the state $|\psi\rangle$, a perfect quantum measurement of Ω yields an eigenvalue ω with probability

$$P(\omega) \propto |\langle \omega|\psi\rangle|^2$$

The state of the system changes:

$$|\psi\rangle \rightarrow |\omega\rangle$$

Note that Ω is Hermitian, so the eigenvalues are real. In terms of the projection operator $\mathbb{P}_\omega = |\omega\rangle\langle\omega|$,

$$\begin{aligned} P(\omega) &\propto |\langle\omega|\psi\rangle|^2 = \langle\psi|\omega\rangle\langle\omega|\psi\rangle \\ &= \langle\psi|\mathbb{P}_\omega|\psi\rangle \\ &= \langle\psi|\mathbb{P}_\omega\mathbb{P}_\omega|\psi\rangle \\ &= \langle\mathbb{P}_\omega\psi|\mathbb{P}_\omega\psi\rangle \end{aligned}$$

4. Time evolution described by

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$H(X, P) = H(x \rightarrow X, p \rightarrow P)$ from the classical Hamiltonian.

Potential problems:

1. $\Omega(x \rightarrow X, p \rightarrow P)$ is ambiguous. E.g. for $\omega = xp$, is $\Omega = XP$ or PX ? Solution is to symmetrize with $\Omega = (XP + PX)/2$.
2. Ω can be degenerate, e.g. two eigenvalues $\omega_1 = \omega_2 = \omega$. Then what is $P(\omega)$? Solution: find some basis $|\omega_1\rangle, |\omega_2\rangle$ in the eigenspace $\mathbb{V}_\omega$. Then

$$P(\omega) = |\langle\omega_1|\psi\rangle|^2 + |\langle\omega_2|\psi\rangle|^2$$

This is the same result we get by assuming ω_1, ω_2 are infinitesimally distinct, then asking for $P(\omega_1 \text{ or } \omega_2)$.

3. Eigenvalue spectrum of Ω can be continuous, e.g. X . Solution: expand

$$|\psi\rangle = \int |\omega\rangle \langle\omega|\psi\rangle d\omega$$

Introduce the **wave function in ω space** $\psi(\omega) = \langle\omega|\psi\rangle$ and the **probability density** $P(\omega) = |\langle\omega|\psi\rangle|^2$. $P(\omega) d\omega$ is the probability of measuring between ω and $\omega + d\omega$. Note $\int P(\omega) d\omega = 1$.

3.1.1 Ensembles

Used to test quantum theory. A quantum ensemble is a large number N of particles all in state $|\omega\rangle$.

A single particle, after measurement, could give any of the eigenvalues. Measure all N to get a statistical result.

Difference with classical:

- In classical ensemble of N particles, if we measure λ_1 in $N/3$ trials and λ_2 in $2N/3$ trials, we expect there were $N/3$ amount of λ_1 and $2N/3$ amount of λ_2 before measurement.
 - In quantum ensemble, all particles are assumed to be $|\omega\rangle$ before measurement.
-

To calculate average over ensemble, use the **expectation value**:

$$\begin{aligned}
 \langle \Omega \rangle &= \sum_i P(\omega_i) \omega_i = \sum_i |\langle \omega_i | \psi \rangle|^2 \omega_i \\
 &= \sum_i \langle \psi | \omega_i \rangle \langle \omega_i | \psi \rangle \omega_i \\
 &= \sum_i \langle \psi | \Omega | \omega_i \rangle \langle \omega_i | \psi \rangle \\
 &= \langle \psi | \Omega | \psi \rangle
 \end{aligned}$$

To measure the uncertainty (fluctuation around the mean), use the **standard deviation**:

$$\Delta \Omega = \langle (\Omega - \langle \Omega \rangle)^2 \rangle^{1/2}$$

Classically, this is called “root-mean-square” deviation. In QM, it’s “uncertainty in Ω .”

$$(\Delta \Omega)^2 = \begin{cases} \sum_i P(\omega_i) (\omega_i - \langle \Omega \rangle)^2 & \text{(discrete spectrum)} \\ \int P(\omega) (\omega - \langle \Omega \rangle)^2 d\omega & \text{(continuous spectrum)} \end{cases}$$

E.g. if a particle has $\langle X \rangle = a$ and $\Delta X = \Delta$, then we know it’s likely to be spotted near a with deviation of order Δ .

3.1.2 Compatible Variables

Is there a filtering process by which we take $|\psi\rangle$ and produce a state with well-defined ω , λ for 2 variables Ω , Λ ?

- Not generally, because measurement changes the state. I.e. (if we do Ω then Λ)

$$\begin{aligned}
 \text{first: } |\psi\rangle &\rightarrow \sum |\omega\rangle \langle \omega | \psi \rangle \\
 \text{second: } |\omega\rangle &\rightarrow \sum |\lambda\rangle \langle \lambda | \omega \rangle
 \end{aligned}$$

But if Ω and Λ have simultaneous eigenkets, then

$$\begin{aligned}
 \Omega |\omega\lambda\rangle &= \omega |\omega\lambda\rangle \\
 \Lambda |\omega\lambda\rangle &= \lambda |\omega\lambda\rangle \\
 (\Omega\Lambda - \Lambda\Omega) |\omega\lambda\rangle &= 0
 \end{aligned}$$

We can “know” both variables at the same time, if it has a simultaneous eigenket. Variables such that $[\Omega, \Lambda] = 0$ are called **compatible**.

E.g. The canonical commutation relation $[X, P] = i\hbar \neq 0$. Thus, a state can’t have well-defined eigenkets of X and P simultaneously. This is the Heisenberg uncertainty relation.

3.1.3 Density Matrices

Instead of an ensemble of N particles all in $|\psi\rangle$, we can consider an ensemble of N systems, n_i of which are $|i\rangle$.

Let $p_i = n_i/N$ be the probability of measuring a $|i\rangle$. Define the density matrix:

$$\rho = \sum_i p_i |i\rangle \langle i|$$

- “Pure”: all $n_i = 0$ except for one
- “Mixed”: otherwise.

The ensemble average of Ω is $\langle \bar{\Omega} \rangle = \sum_i p_i \langle i | \Omega | i \rangle$. Note that

$$\begin{aligned} \text{Tr}(\Omega \rho) &= \sum_j \langle j | \Omega \rho | j \rangle = \sum_i \sum_j \langle j | \Omega \rho | i \rangle \langle i | j \rangle p_i \\ &= \sum_i \sum_j \langle i | j \rangle \langle j | \Omega \rho | i \rangle p_i = \sum_i \langle i | \Omega \rho | i \rangle p_i \\ &= \langle \bar{\Omega} \rangle \end{aligned}$$

E.g. if we want the probability of obtaining ω , $\overline{P(\omega)}$, then

$$\begin{aligned} P(\omega) &= |\langle \omega | \psi \rangle|^2 = \langle \psi | \omega \rangle \langle \omega | \psi \rangle = \langle \psi | \mathbb{P}_\omega | \psi \rangle \\ \overline{P(\omega)} &= \text{Tr}(\mathbb{P}_\omega \rho) \end{aligned}$$

Features:

- $\rho^\dagger = \rho$
- $\rho^2 = \rho$ for pure state
- $\rho = \frac{1}{k} I$ for mixed state uniformly distributed over k sub-states
- $\text{Tr}(\rho) = 1$
- $\text{Tr}(\rho^2) \leq 1$, equality if ρ is pure.

3.2 Example: Gaussian Wavefunction

We'll find the Heisenberg uncertainty relation (sort of) for this model.

Put the wavefunction in the X -basis:

$$|\psi\rangle = \int_{-\infty}^{\infty} |x\rangle \langle x | \psi \rangle dx = \int_{-\infty}^{\infty} |x\rangle \psi(x) dx$$

Assume $\psi(x)$ is Gaussian and normalize to unity:

$$\psi(x) = \frac{1}{(\pi \Delta^2)^{1/4}} e^{-(x-a)^2/(2\Delta^2)}$$

In the X -basis, the effect of X is to multiply by x . I.e.

$$\begin{aligned}\langle x|X|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|X|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} x\delta(x-x')\psi(x') dx' \\ &= x\psi(x)\end{aligned}$$

Expectation value is $\langle X \rangle = a$ (which makes sense; $\psi(x)$ is centered around a).

Also, $\langle X^2 \rangle = \Delta^2/2 + a^2$

Then the uncertainty is

$$\begin{aligned}\Delta X &= \left[\langle \psi | (X - \langle X \rangle)^2 | \psi \rangle \right]^{1/2} = \left[\langle X^2 \rangle - 2a\langle X \rangle + a^2 \right]^{1/2} \\ &= \left[\langle X^2 \rangle - a^2 \right]^{1/2} \\ &= \frac{\Delta}{\sqrt{2}}\end{aligned}$$

Now for P . The action in the X basis is $-i\hbar \frac{d}{dx}$. I.e.

$$\begin{aligned}\langle x|P|\psi\rangle &= \int_{-\infty}^{\infty} \langle x|P|x'\rangle \langle x'|\psi\rangle dx' = \int_{-\infty}^{\infty} -i\hbar\delta'(x-x')\psi(x') dx' \\ &= -i\hbar \frac{d\psi}{dx}\end{aligned}$$

Now find the eigenvalues. Note that $\langle x|p\rangle = \psi_p(x) \neq \psi(x)$.

$$\begin{aligned}P|p\rangle &= p|p\rangle \longrightarrow \langle x|P|p\rangle = p\langle x|p\rangle \\ -i\hbar \frac{d\psi_p(x)}{dx} &= p\psi_p(x) \\ \psi_p(x) &= \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}\end{aligned}$$

By finding $\psi(p) = \langle p|\psi\rangle$, we get a Gaussian of width $\hbar/(\Delta\sqrt{2})$ centered at 0. Thus $\langle P \rangle = 0$ and $\Delta P = \hbar/(\Delta\sqrt{2})$.

Put together: for this Gaussian wavefunction model, we get

$$\Delta X \cdot \Delta P = \frac{\hbar}{2}$$

This is a lower bound on the Heisenberg uncertainty relation.

3.3 Generalization to N Degrees of Freedom

Use mutually commuting operators $X_1, \dots, X_N$. In their simultaneous eigenbasis $|x_1, x_2, \dots, x_N\rangle$,

$$\langle x_1, x_2, \dots, x_N | x'_1, x'_2, \dots, x'_N \rangle = \delta(x_1 - x'_1) \cdots \delta(x_N - x'_N)$$

And

$$\begin{aligned} |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | \psi \rangle = \psi(x_1, \dots, x_N) \\ X_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | X_i | \psi \rangle = x_i \psi(x_1, \dots, x_N) \\ P_i |\psi\rangle &\longrightarrow \langle x_1, \dots, x_N | P_i | \psi \rangle = -i\hbar \frac{\partial}{\partial x_i} \psi(x_1, \dots, x_N) \end{aligned}$$

Then follow the procedure of replacing classical variables with quantum operators. E.g. for the classical operator

$$\omega = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + x^2 + y^2 + z^2$$

We associate the quantum operator such that (with $X_1 = X$, $X_2 = Y$, and $X_3 = Z$)

$$\Omega |\omega\rangle = \omega |\omega\rangle \quad \Omega = \frac{P_x^2 + P_y^2 + P_z^2}{2m} + X^2 + Y^2 + Z^2$$

Then substitute $|\omega\rangle = \psi_\omega(x, y, z)$, $X \rightarrow x$, and $P_x \rightarrow -i\hbar \frac{d}{dx}$.

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + x^2 + y^2 + z^2 \right] \psi_\omega(x, y, z) = \omega \psi_\omega(x, y, z) \quad (1)$$

Once we have this form, we can change to other coordinates, e.g. spherical. Use

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \equiv \nabla^2 \equiv \frac{1}{r^2} \left(\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

And substitute into (1) to get

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi_\omega}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi_\omega}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi_\omega}{\partial \phi^2} \right] + r^2 \psi_\omega = \omega \psi_\omega \quad (2)$$

3.4 The Schrödinger Equation

3.4.1 Hamiltonian Setup

Simply substitute $x \rightarrow X$, $p \rightarrow P$ in the classical Hamiltonian. E.g. for (1) harmonic oscillator, (2) constant force f , and (3) particle of charge q in an EM field,

$$\begin{aligned} 1) \quad H &= \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 &\longrightarrow H &= \frac{P^2}{2m} + \frac{1}{2} m \omega^2 X^2 \\ 2) \quad H &= \frac{p^2}{2m} - f x &\longrightarrow H &= \frac{P^2}{2m} - f X \\ 3) \quad H &= \frac{|\vec{p} - \frac{q}{c} \vec{A}(\vec{r}, t)|^2}{2m} + q\phi(\vec{r}, t) &\longrightarrow H &= \frac{1}{2m} \left(P \cdot P - \frac{q}{c} A \cdot P - \frac{q}{c} P \cdot A + \frac{q^2}{c^2} A \cdot A \right) + q\phi \end{aligned}$$

Note: P doesn't commute with A (which depends on X, Y, Z).

3.4.2 Time-Independent Schrödinger Equation

Start from Postulate IV, $i\hbar \frac{\partial}{\partial t} |\psi\rangle = H |\psi\rangle$. We seek to find a propagator of the form

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle$$

Use the energy basis, since $H |E\rangle = E |E\rangle$.

$$|\psi(t)\rangle = \sum_E |E\rangle \langle E|\psi(t)\rangle \equiv \sum_E a_E(t) |E\rangle$$

Apply $(i\hbar \frac{\partial}{\partial t} - H)$ and use $(i\hbar \frac{\partial}{\partial t} - H) |\psi(t)\rangle = 0$ to get

$$\begin{aligned} 0 &= \sum_E (i\hbar \frac{\partial}{\partial t} - H) a_E(t) |E\rangle = \sum_E (i\hbar \dot{a}_E - E a_E) |E\rangle \\ a_E(t) &= a_E(0) e^{-iEt/\hbar} \\ \langle E|\psi(t)\rangle &= \langle E|\psi(0)\rangle e^{-iEt/\hbar} \end{aligned}$$

Now plug back in to $|\psi\rangle$ to get the propagator:

$$\begin{aligned} |\psi(t)\rangle &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E|\psi(0)\rangle \\ U(t) &= \sum_E e^{-iEt/\hbar} |E\rangle \langle E| = e^{-iHt/\hbar} \end{aligned} \tag{1}$$

Because H is unitary, $U(t)$ is unitary, thus time evolution is unitary and the norm is preserved: $\langle \psi(t)|\psi(t)\rangle = \langle \psi(0)|\psi(0)\rangle$.

3.4.3 Time-Dependent Schrödinger Equation

To approximate a time-dependent Hamiltonian, split $(0, t)$ into N intervals of width $\Delta = t/N$. Then

$$\begin{aligned} |\psi(t)\rangle &= |\psi(0)\rangle + \Delta \left. \frac{d|\psi\rangle}{dt} \right|_0 = |\psi(0)\rangle - \frac{i\Delta}{\hbar} H(0) |\psi(0)\rangle \\ &= \left[1 - \frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \\ &\approx \exp \left[-\frac{i\Delta}{\hbar} H(0) \right] |\psi(0)\rangle \end{aligned}$$

The overall propagator is found by summing the changes:

$$\begin{aligned} |\psi(t)\rangle &= \prod_{n=0}^{N-1} e^{-i\Delta H(n\Delta)/\hbar} |\psi(0)\rangle \\ U(t) &= \exp \left[-\frac{i}{\hbar} \int_0^t H(t') dt' \right], \quad N \rightarrow \infty \end{aligned} \tag{2}$$

4 Simple Problems in 1D

4.1 The Free Particle

4.1.1 Allowed Energy States

Start from the Schrodinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle = \frac{P^2}{2m} |\psi\rangle$$

The normal modes (in energy basis) are

$$|\psi\rangle = e^{-iEt/\hbar} |E\rangle$$

We seek to find allowed E states. $|p\rangle$ is an eigenvector of both P and P^2 , so

$$\begin{aligned} \frac{P^2}{2m} |p\rangle = E |p\rangle &\longrightarrow \frac{p^2}{2m} |p\rangle = E |p\rangle \\ p &= \pm\sqrt{2mE} \end{aligned}$$

Thus for each eigenvalue E , we have 2 eigenvectors:

$$\begin{aligned} |E, +\rangle &= |p = \sqrt{2mE}\rangle \\ |E, -\rangle &= |p = -\sqrt{2mE}\rangle \end{aligned}$$

I.e. a particle with energy E may be found moving left or right with momentum $\sqrt{2mE}$. So far, this agrees with a classical analysis. The difference is that in QM, we can have a state like

$$|E\rangle = \beta |p = \sqrt{2mE}\rangle + \gamma |p = -\sqrt{2mE}\rangle$$

4.1.2 Propagator

Use the propagator from the time-independent Schrodinger equation:

$$U(t) = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-iE(p)t/\hbar} dp = \int_{-\infty}^{\infty} |p\rangle \langle p| e^{-ip^2t/2m\hbar} dp$$

Note that in the X basis, the propagator for a time slice is

$$\begin{aligned} U(x, t; x') &\equiv \langle x|U(t)|x'\rangle = \int_{-\infty}^{\infty} \langle x|p\rangle \langle p|x'\rangle e^{-ip^2t/2m\hbar} dp \\ &= \frac{1}{2m\hbar} \int_{-\infty}^{\infty} e^{ip(x-x')/\hbar} \cdot e^{-ip^2t/2m\hbar} dp \\ &= \sqrt{\frac{m}{2\pi\hbar it}} e^{im(x-x')^2/2\hbar t} \end{aligned}$$

In terms of the propagator, given the initial state $\psi(x', 0)$, we can find an arbitrary state $\psi(x, t)$ as

$$\psi(x, t) = \int U(x, t; x') \psi(x', 0) dx'$$

4.2 The Particle in a Box

4.2.1 Wavefunction

Consider a box potential:

$$V(x) = \begin{cases} 0 & |x| < L/2 \\ \infty & |x| \geq L/2 \end{cases}$$

Let I and III be the regions outside the box, while II is the region within the box. Use the eigenvalue equation in the X basis:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

In III, we have

$$\begin{aligned} \frac{d^2\psi_{\text{III}}}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2}\psi_{\text{III}} &= 0, \quad V_0 \rightarrow \infty \\ \psi_{\text{III}} &= Ae^{-\kappa x} + Be^{\kappa x}, \quad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \end{aligned}$$

We must have $B = 0$ because otherwise the wavefunction blows up as $x \rightarrow \infty$. Then as $V_0 \rightarrow \infty$, $\psi_{\text{III}} = 0$. This is as expected. Similar results hold for I: $\psi_{\text{I}} = 0$.

In II, the solution is that of a free particle:

$$\psi_{\text{II}} = Ae^{ikx} + Be^{-ikx}, \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

Further, it must obey the boundary conditions:

$$\begin{aligned} \psi_{\text{I}}(-L/2) = \psi_{\text{II}}(-L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{ikL/2} = 0 \\ \psi_{\text{III}}(L/2) = \psi_{\text{II}}(L/2) = 0 &\longrightarrow Ae^{ikL/2} + Be^{-ikL/2} = 0 \\ \begin{bmatrix} e^{-ikL/2} & e^{ikL/2} \\ e^{ikL/2} & e^{-ikL/2} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

For this to happen, the determinant must be 0, i.e. $e^{ikL} - e^{-ikL} = 0$, thus $k = n\pi/L$, where $n = 0, \pm 1, \pm 2, \dots$. We can find that $A = -e^{in\pi}B$, leading to two families of solutions:

$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) & n \text{ even} \\ \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right) & n \text{ odd} \end{cases}$$

4.2.2 Allowed Energies

Note that the energy is quantized:

$$E = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

And further, the lowest bound on energy (for non-trivial solutions) is nonzero: the ground state energy E_g is

$$E_g = \frac{\hbar^2 \pi^2}{2mL^2}$$

An “intuitive” explanation for this is that Δx is bounded in the well, so by the uncertainty principle, we cannot have a well-defined momentum of 0.

The uncertainty principle also provides a quick estimate of the ground energy. Start with $H = P^2/2m$. From the Ehrenfest theorem, $\langle H \rangle = \langle P^2 \rangle / 2m$. Since $\langle P \rangle = 0$ inside the box, we can further write

$$\langle H \rangle = \frac{\langle (P - \langle P \rangle)^2 \rangle}{2m} = \frac{(\Delta P)^2}{2m}$$

From the uncertainty relations, $\Delta X \cdot \Delta P \geq \hbar/2$, so we have

$$\langle H \rangle \geq \frac{\hbar^2}{8m(\Delta x)^2} \geq \frac{\hbar^2}{2mL^2}$$

In energy eigenstates, $\langle H \rangle = E$. Thus we have

$$E \geq \frac{\hbar^2}{2mL^2}$$

Which is the estimated order of the ground state, accurate up to a factor of π^2 .

4.3 The Continuity Equation of Probability

From classical electrodynamics:

$$\text{(global)} \quad Q_{\text{total}} = \text{const.}$$

$$\text{(local)} \quad \frac{d}{dt} \int_V \rho(\mathbf{r}, t) d^3\mathbf{r} = - \int_{S_V} \mathbf{j} \cdot d\mathbf{S}$$

In QM:

$$\text{(global)} \quad \iiint P(\mathbf{r}, t) d^3\mathbf{r} = \text{const.}$$

$$\text{(local)} \quad \frac{d}{dt} \int P(\mathbf{r}, t) d^3\mathbf{r} = - \int_{S_\infty} \mathbf{j} \cdot d\mathbf{S} \quad , \quad \mathbf{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

In a nutshell, the total probability to find a particle anywhere in the universe is conserved. Locally, the reduction in probability of a region must correspond to a change in the probability current $\mathbf{j}$ is the **probability current density**.

4.4 Quantum Tunneling

The details are complicated, see Shankar chp. 5 for more. The main idea: instead of an infinite potential, we have a region of 0 potential I, and a region of *finite* potential II. Let $V = 0$ in I and $V = V_0$ in II.

$$\begin{aligned} \text{I: } \psi_E(x) &= Ae^{ik_1x} + Be^{-ik_1x} \quad , \quad k_1 = \sqrt{\frac{2mE}{\hbar^2}} \\ \text{II: } \psi_E(x) &= Ce^{ik_2x} + De^{-ik_2x} \quad , \quad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \end{aligned}$$

Set $D = 0$ for the right-traveling transmitted waves. At $x = 0$, boundary conditions require that

$$\begin{aligned} A + B &= C \\ ik_1(A - B) &= ik_2C \end{aligned}$$

After some calculations, we find the reflection probability (R) and transmission probability (T) to be:

$$\begin{aligned} R &= \left| \frac{B}{A} \right|^2 = \left| \frac{\sqrt{E_0} - \sqrt{E_0 - V_0}}{\sqrt{E_0} + \sqrt{E_0 - V_0}} \right|^2 \\ T &= 1 - R = \left| \frac{4\sqrt{E_0}\sqrt{E_0 - V_0}}{(\sqrt{E_0} + \sqrt{E_0 - V_0})^2} \right| \end{aligned}$$

The key is that even if $E_0 < V_0$ (energy not enough to overcome barrier), there's still a nonzero probability for the particle to be transmitted. This is called **quantum tunneling**.

4.5 Ehrenfest's Theorem

QM applied to many particles should reproduce the macroscopic world.

Consider the time evolution of expectation values:

$$\frac{d}{dt} \langle \Omega \rangle = \frac{d}{dt} \langle \psi | \Omega | \psi \rangle = \langle \dot{\psi} | \Omega | \psi \rangle + \langle \psi | \Omega | \dot{\psi} \rangle + \langle \psi | \dot{\Omega} | \psi \rangle$$

Assume Ω has no time dependence, so $\dot{\Omega} = 0$. Further, from the Schrodinger equation,

$$\begin{aligned} |\dot{\psi}\rangle &= -\frac{i}{\hbar} H |\psi\rangle \\ \langle \dot{\psi}| &= \frac{i}{\hbar} \langle \psi| H \end{aligned}$$

Combining the two results gives **Ehrenfest's theorem**:

$$\frac{d}{dt} \langle \Omega \rangle = -\frac{i}{\hbar} \langle [\Omega, H] \rangle$$

Note the similarity with classical mechanics:

$$\frac{d\omega}{dt} = \{\omega, H\}$$

For a Hamiltonian of the form $H = P^2/2m + V(X)$, we can also get similar forms to the canonical equations:

$$\langle \dot{X} \rangle = \left\langle \frac{\partial H}{\partial P} \right\rangle \quad , \quad \langle \dot{P} \rangle = \left\langle -\frac{\partial H}{\partial X} \right\rangle$$

Thus we can regain classical physics if the fluctuation about the mean is small. But the above equations depend on the choice of H .